

IMPROVING THE EFFICACY OF VSLAM IN DYNAMIC ENVIRONMENTS

Introduction

- Autonomous robots are increasingly deployed in human-centered environments
- Reliable **localization** and **navigation** are required to ensure **human safety** and **robot reliability**
- **Visual Simultaneous Localization and Mapping (VSLAM)** enables **autonomous navigation** by estimating camera pose and constructing a 3d map of the surrounding environment
- Many common approaches rely on feature correspondences for tracking and mapping, which utilize **static landmarks**
- **Dynamic objects** (i.e. people, vehicles, animals) introduce dynamic features
- These features cause **pose estimation errors** and **trajectory drift**, reducing localization accuracy and accurate 3d map reconstruction
- Improving **robustness** of **VSLAM** algorithms in **dynamic** environments remains an active research **challenge**

Purpose

- Investigate whether **semantic filtering** of dynamic pixels can improve VSLAM performance in **dynamic environments**
- Utilize a **YOLO-based segmentation model** to the VSLAM pipeline to **exclude dynamic features** prior to tracking

Methodology

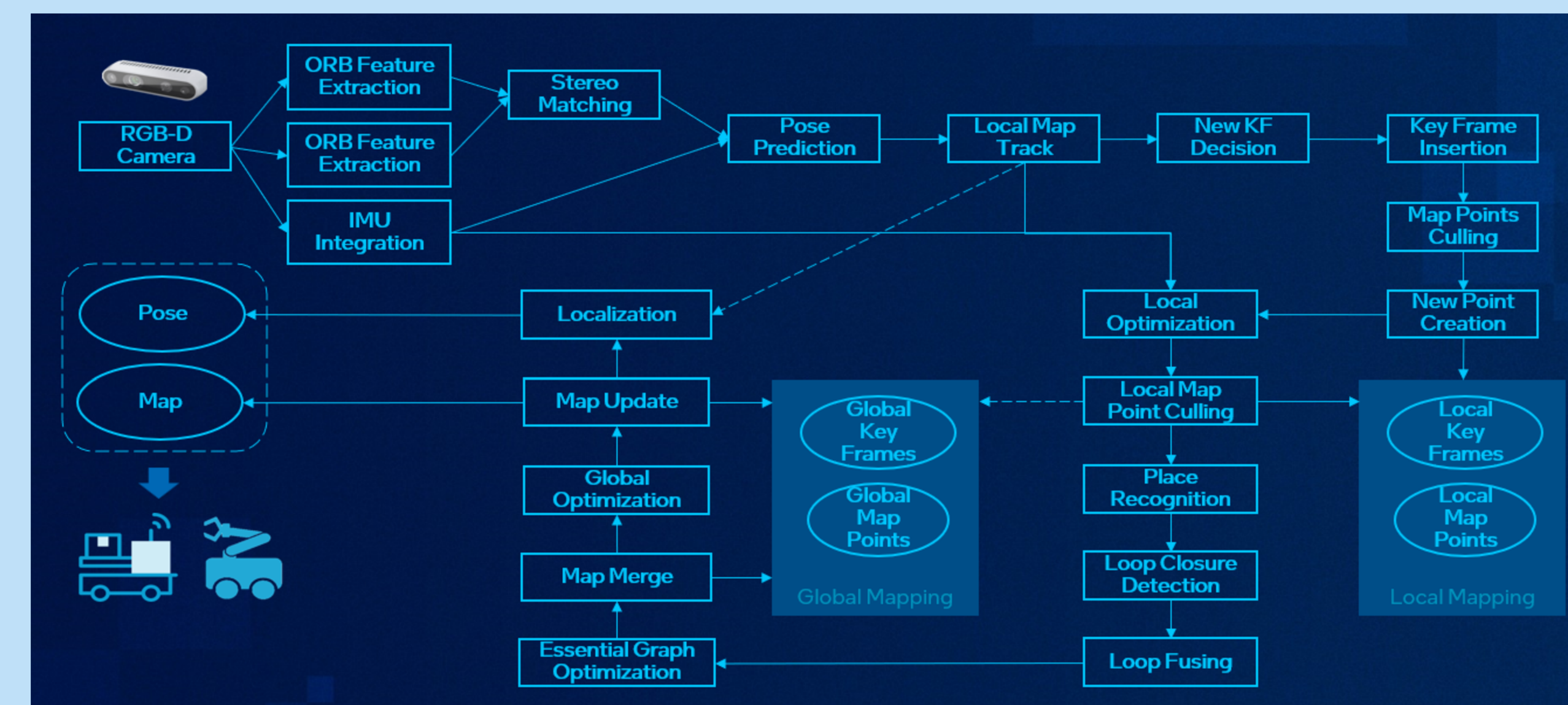


Figure 1. Graph of ORBSLAM3 Pipeline (Intel, n.d.)



Figure 2. Semantic Segmentation on a Sample Image (Towards AI Editorial Team, 2021)

- **YOLOv8-seg** is utilized as a semantic segmentation network to **classify individual pixels** as **dynamic** or **static**
- Features derived from **dynamic pixels** are **discarded**

Performance was evaluated via **Absolute Trajectory Error (ATE)** to measure global trajectory error, **Relative Pose Error (RPE)** to measure local motion consistency, and **median tracking time** to evaluate computational load.

Semantic filtering was added to the STELLA VSLAM implementation of ORB SLAM-3

Results

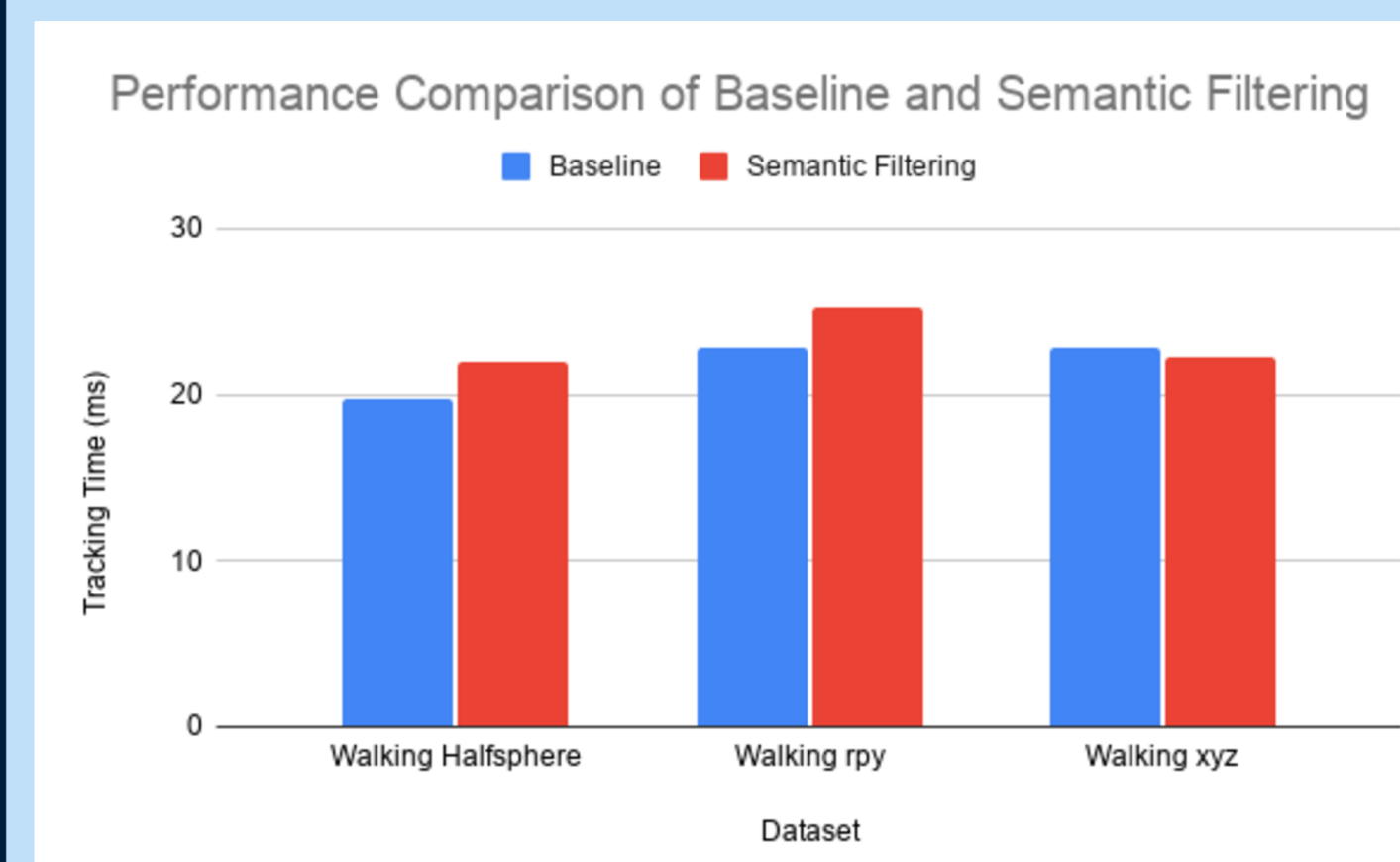


Figure 3. Graph of Performance Comparison created by researcher using Google Sheets

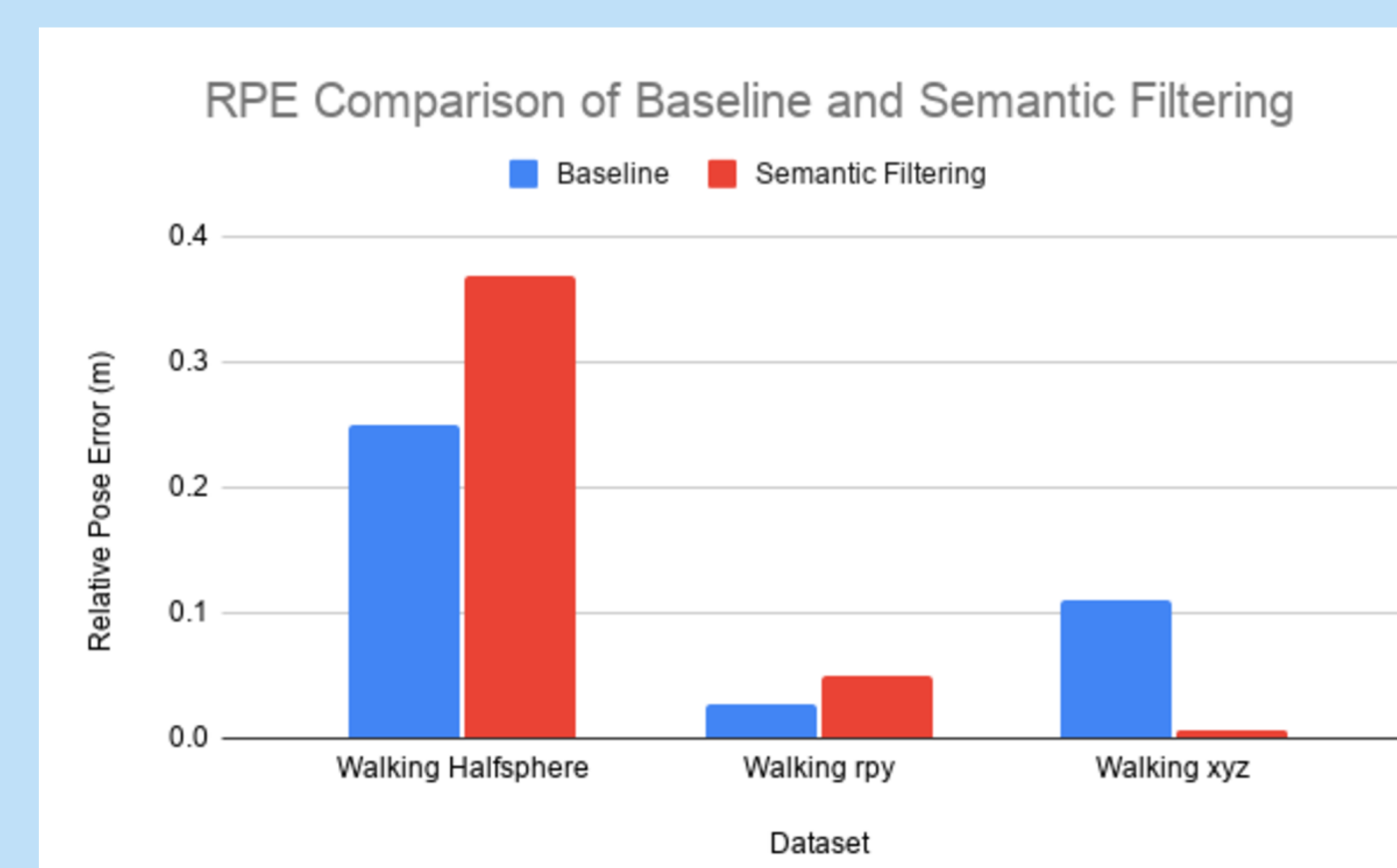


Figure 4. Graph of RPE Comparison created by researcher using Google Sheets

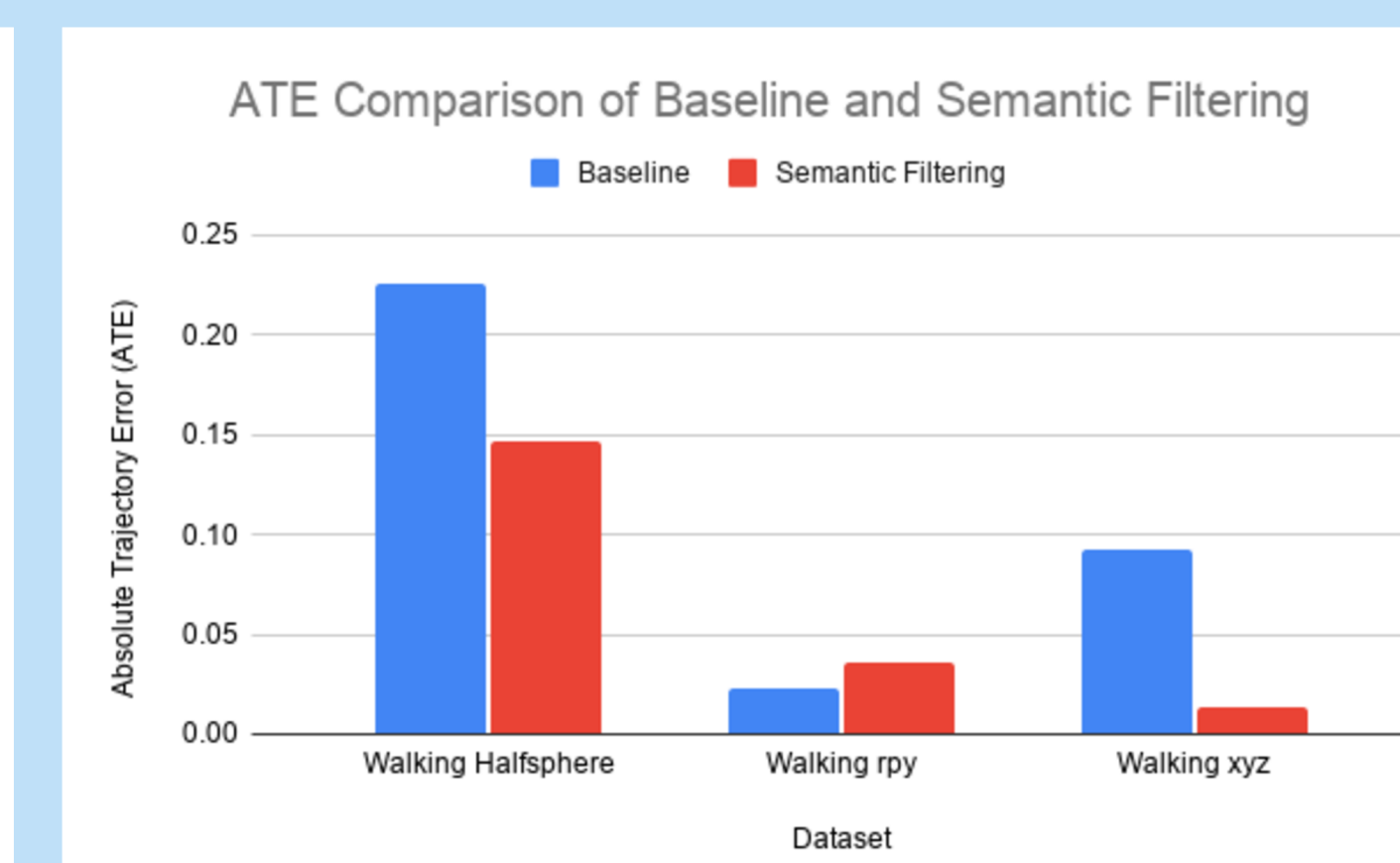


Figure 5. Graph of ATE Comparison created by researcher using Google Sheets

The modified algorithm with semantic filtering was compared with the baseline algorithm on 3 datasets with varying types of motion (Halfsphere, RPY, XYZ).

Conclusion

- ATE improvement suggests excluding dynamic features reduces long-term trajectory drift
- Increase in RPE suggests mild short term pose instability
- **Semantic filtering** improved global localization accuracy in dynamic environments
- Enhances the reliability of robots navigating in environments with multiple moving objects

Future Work

- Dynamic objects contain large amounts of features, which may remain stable across frames
- Filtering removes **all** features, impacting localization accuracy
- Future work will investigate **geometric filtering** to identify individual features that do not maintain **consistent geometric relationships**
- Selectively exclude **unstable** features, improving **robustness**
- May further reduce ATE while maintaining RPE

References

- Intel. (n.d.). *VSLAM: ORB SLAM3*. Open Edge Platform. Retrieved 2026, from https://docs.openedgeplatform.intel.com/dev/edge-ai-suites/robotics-ai-suite/embodied/sample_pipelines/ORB_VSLAM.html
- Towards AI Editorial Team. (2021, October 21). *Semantic Segmentation: A Complete Guide*. Towards AI. Retrieved 2026, from <https://towardsai.net/p/ll-machine-learning-7>
- Sumikura, S., Shibuya, M., & Sakurada, K. (2019). OpenVSLAM: A versatile visual SLAM framework. Proceedings of the 27th ACM International Conference on Multimedia. <https://doi.org/10.1145/3343031.3350539>